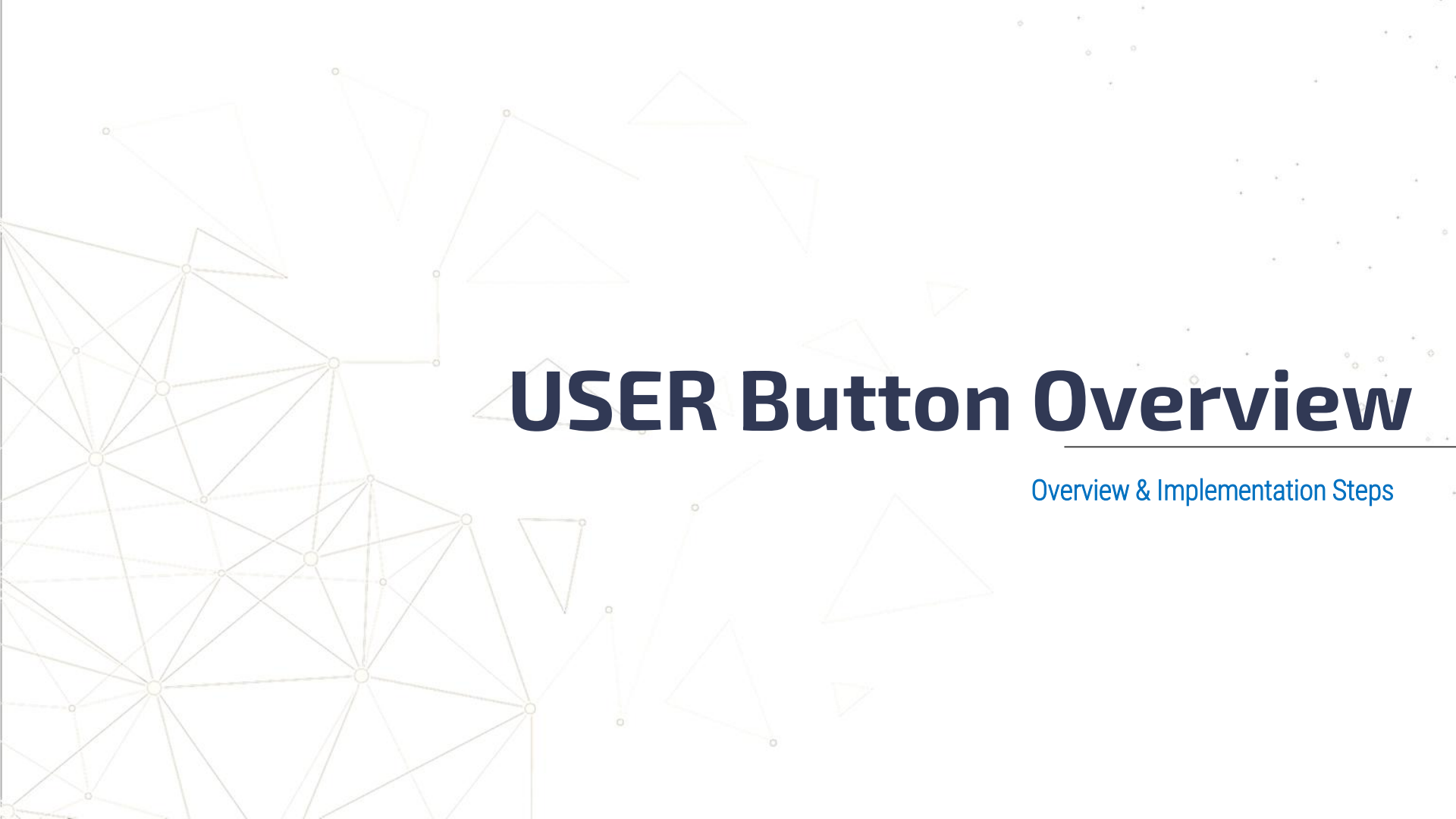


The background features a complex network of thin, light-colored lines and dots, forming a web-like structure. Overlaid on this are several translucent triangles of various sizes and orientations, some with solid outlines and others with dashed or dotted lines. The overall aesthetic is technical and modern.

USER Button Overview

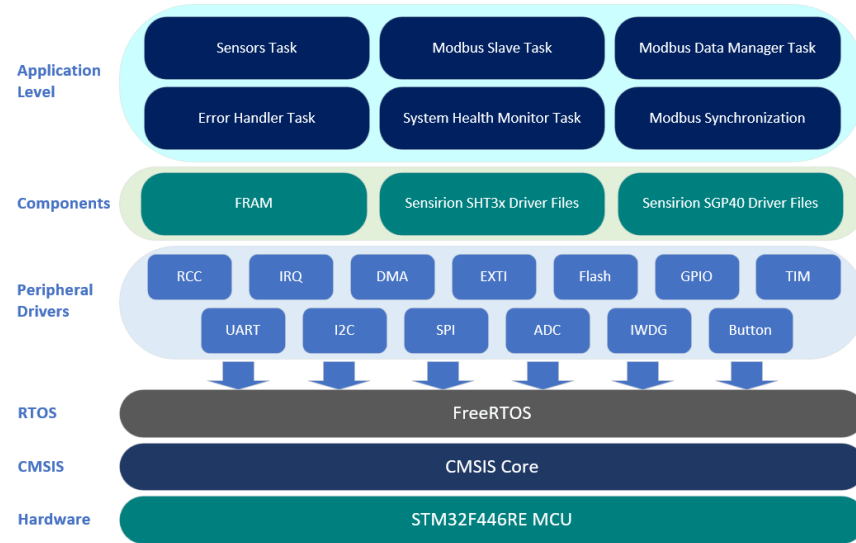
Overview & Implementation Steps

USER Button Overview

- USER Button Overview
- USER Button Implementation
- Button Driver Overview
- Practical Implementation Steps

USER Button Overview

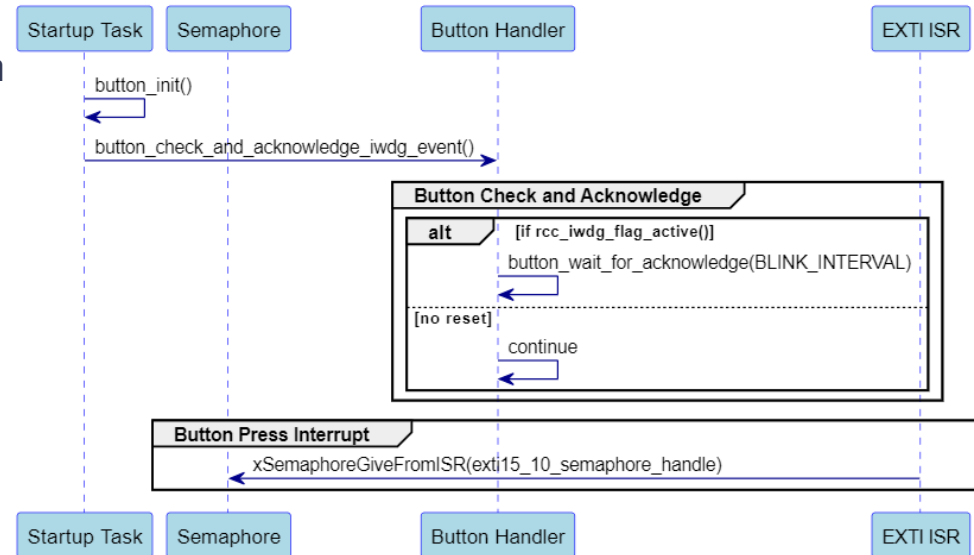
- **Definition:** The USER button is a GPIO input on the Nucleo board connected to Port C, Pin 13.
- **Purpose & Functionality:**
 - **Event Acknowledgement:** Used to acknowledge an IWDG reset. During system startup, if an IWDG reset is detected, the system halts and blinks an LED to alert the user, resuming normal operation only after the button is pressed.
- **Project Significance:**
 - **User Button Acknowledgement:** Useful for USER awareness of IWDG events and for debugging during development.



USER Button Implementation

Key Points

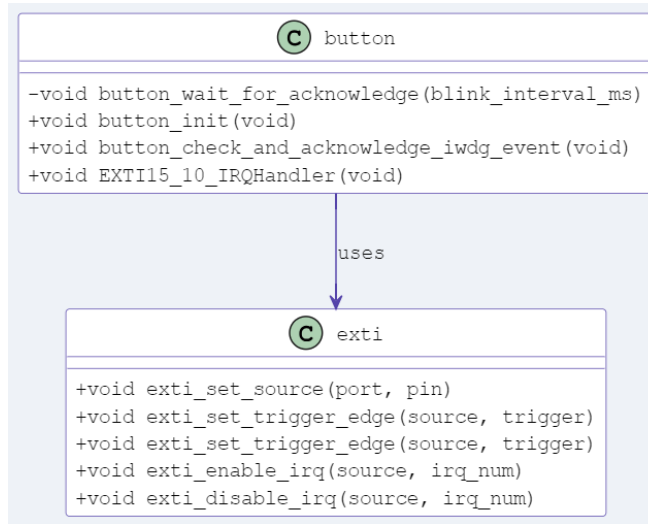
- **GPIO Initialization:** Configured as a GPIO Input in `gpio.c`. On Nucleo Boards, the line is pulled low when the button is pressed.
- **USER Button Initialization:** Creates a binary semaphore for EXTI synchronization and configures EXTI line with falling edge trigger interrupt.
- **Button Handler:** Uses the Semaphore to synchronize user acknowledgment.
- **EXTI ISR:** `EXTI15_10_IRQHandler` Gives Semaphore upon button press allowing the system to proceed.



Button Driver Overview

Functions

- **Wait for Acknowledge:** Blinks an LED until the user presses the button to acknowledge an IWDG reset event. Uses [xSemaphoreTake](#).
- **Initialization:** Sets up the button by creating a semaphore and configuring the EXTI line for interrupts. Uses [xSemaphoreCreateBinary](#).
- **Check & Acknowledge IWDG Event:** Verifies if an IWDG reset occurred and waits for user acknowledgment via the button press using the ``button_wait_for_acknowledge()``.
- **EXTI15_10_IRQHandler:** Handles the EXTI interrupt triggered by the button press, giving the semaphore to resume operation. Uses [xSemaphoreGiveFromISR](#).



Practical Implementation Steps

Key Points

- **GPIO Configuration:** Configure the USER Button as a GPIO Input. This configuration allows the microcontroller to detect the button press as a digital signal.
- **Button Header File:** Define the blinking interval and public function prototypes.
- **Button Source File:** Implement functions to initialize the button, handle interrupts, and manage user acknowledgment.

Importance

- **User Acknowledgement:** Provides a safety measure by halting the application until USER acknowledgement. This is critical for ensuring system reliability and aids in debugging during development.

The background features a complex network of thin, light-colored lines and dots, forming various triangular shapes. Some triangles are filled with a very light yellow or green color, while others are just outlines. The overall effect is a subtle, geometric pattern.

Next: Button Header File
